Foundation for Virtual and Augmented Reality Systems

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Lecture – 22

Lec-22: AR Software

Hello and welcome to the NPTEL MOOCs course on Virtual and Augmented Reality Systems. We are in lecture number 11 where we are discussing AR software. We will continue our discussion on AR software. So, earlier we talked about the basic categories of AR software. We learned about different ways to categorize and different ways to view AR software.

One such category is based on the skill requirement from the part of the developer. Another categorization can be based on the platform, the hardware, the operating system, or a combination of both, or even mobile versus desktop platforms. Mobile versus wearable platforms. So, different ways to categorize. Then we have also seen that it is possible to categorize based on technology used, particularly on tracking.

We are following the skill requirement-based categorization in this discussion. So, we have covered low-level libraries and frameworks, which is one of the categories. Then we started our discussion on the second category, which is rapid prototyping tools. We partly covered that; we are going to continue our discussion on that category, and also we are going to cover the other categories in this particular way of looking at AR software. So, let us continue our discussion with rapid prototyping tools.

So, earlier we talked about flash-based tools. Let us see non-flash-based tools, if there are any. So, we mentioned that it is possible to avoid, I mean it is possible to have non-flash-based tools. Now, it is also possible to avoid third-party browser plugins altogether. So, earlier tools that we talked about were primarily meant for browser-based experiences.

where we require plugins, say for example flash-based plugins or non-flash system-based plugins. But when we are talking about other tools, we can say that it is also possible to avoid third-party browser plugins altogether. Instead, you can build your own using HTML, WebGL, or other such web component technologies. You can write your own program. If you have mastery over such technologies, then you don't need to use any third-party plugin.

Instead, you can go for writing your own code to create a fast prototype. There are also non-web-based authoring tools meant for mobile and desktop platforms. So both environments are supported with such tools. For example, Open Frameworks. As the name suggests, this is a framework which is a combination of multiple libraries.

It is meant for desktop and mobile platforms. Now, Open Frameworks is quite widely used. It is an open-source C++ middleware library. With Open Frameworks, it is easy to build interactive experiences while building an AR application. As the name suggests, and we have already mentioned, it provides a framework for combining a wide range of different interactive technologies.

So, the framework provides support for tracking, rendering, as well as interaction. All three are supported. For example, An AR application could be developed by combining a tracking library such as AR Toolkit with hand tracking, a 3D scene graph for rendering, and audio playback libraries. So, we can combine more than two libraries for generating different effects in our proposed application.

For mobile applications, one useful prototyping tool is Wikitude. Typically, such tools are AR browser applications. Wikitude is one such example. Now, these are free applications.

They run on mobile devices. The computation does not take place on the mobile device. Instead, they connect back to some servers for providing content and specifying user interface elements on mobile. So, mobile is primarily treated as a client device or as a rendering device only; actual computation may take place on a backend server, not on the mobile. So if the developers have their own content, then it will be easier or faster to prototype the ideas.

If the content is not there, the developer has to depend on some default content provided by such tools, which may reduce some sort of flexibility, but still it can be used. That is about the second category. Now, let us move our attention to the third category, that is, plugins to existing tools. Partly we have talked about this in the previous category as well.

Here the idea is that when we are talking about software libraries, they require programming skills to use them.

You have to first understand and then use them. So, one has to learn or master the libraries. As opposed to that, now suppose that one is already aware of some interactive 2D or 3D content authoring tool, some tool that the user is already aware of, which may not be relevant at this stage for AR or VR system development or prototype development, but it is relevant for 2D or 3D content authoring. So, in that case, the user does not need to understand or master a new tool completely from scratch.

What the user or developer can do is use plugins to existing such software. These plugins add AR-specific functionalities to the tools. So, the tools the user may be aware of which may not support AR, Content development or your system development whereas plugins if added to the known tool can provide that support. So, the developer does not need to learn everything from scratch instead the developer can rely on the knowledge of that original tool and the developer needs to understand the plugin only.

The requirement of understanding the system thus reduces. So, plugins are therefore useful when one already knows how to use the authoring tool with which the plugin is being added. If the user does not know the authoring tool, then of course it makes no sense; the user has to learn everything from scratch. As we have already mentioned, plugins simply add, some AR visualization and tracking functions to the existing authoring tool and it relies on functions provided by the main authoring tool for content authoring.

So, if you know the original tool then you can create your own content and then use the AR plugin for adding some functionalities related to tracking related to visualization of or 3D registration of digital content over real world view. Some example plugins, let us see. There are actually many interactive 3D graphics authoring tools to create interactive 3D applications such as Unity, Vizard, Lumion, Now, most of these tools have their own plugins for AR visualization.

So, they come with AR-based plugins. The Unity tool has the widest range of solutions available for AR visualization. For example, AR Toolkit for Unity from the AR Tool Works can be one such plugin. It can provide both marker and image-based or computer vision-based tracking and AR visualization support. Similarly, the Vuforia tracking library has an AR plugin that works with Unity 3D.

So, Vuforia can be used as a plugin with Unity 3D to create AR applications, particularly mobile AR applications for both iOS and Android platforms. So what are the pros? We have already discussed when we are using a plugin-based approach, plugin-based tools. So they provide an easy and quick method for developing AR applications. But there are cons, there is, I mean, a negative side also.

User interface and interaction design can be limited to those provided by the authoring tool. So the underlying authoring tool provides such support whereas plugins provide AR support. Now the support provided by the underlying authoring tool restricts the developer's flexibility in deciding on different interaction methods or interface look and feel. Software and hardware configuration also depends on what is provided by the tool.

So again, flexibility is lost. Content produced with the authoring tool usually becomes dependent on proprietary player software. So if you have some multimedia content, then to play that content, you have to, in most cases, not always, in most cases, you may have to depend on proprietary players. And that again reduces some sort of flexibility. Now, these are cons.

So, there is a positive side, there is a negative side. And you know, as we all know, any approach will have both sides. In this case, if we wish to resolve those issues, those cons or the negative sides, what we can do is we can rely on the hard way, that is we can use low-level software development tools, such as software libraries and build our own system from scratch rather than depending on some third-party tool. That is applicable for any higher-level tool as such.

Now, let us move to the last category, that is standalone authoring tools. So, here the idea is that standalone applications are there that allow users to create their own AR experiences with no programming experience. So, even if someone does not have any programming experience, he or she can still be able to build an AR application using such authoring tools. Unlike plugins, these tools do not require any additional software to work.

So, plugins require A basic software, a basic authoring tool to work on which AR support will be provided through a plugin. In this case, they are standalone; they come as a complete package with content authoring as well as AR support. So, they do not require or depend on any other tool. And these authoring tools provide different sets of functions, as is obvious.

So, they support scene construction and animation, and also support adding interactive behavior. So, these are typically common function sets that are supported by such standalone authoring tools. Some of these tools only provide scene construction functions, while others have the full capability of building complex interactive behavior. So, they come with different strengths. come in different strengths; some provide limited support, some provide extensive support, which is again quite common in different software packages now, most of these tools come with some sort of graphical user interface for the developer to create an application, create an experience. Now, these GUIs are drag-and-drop type. So, they allow drag-and-drop type interaction with the GUI to create an experience. In that sense, they do not require—the developers do not require any coding as such. But for flexibility, some sort of coding support, if the developer knows it, can be provided using scripting languages.

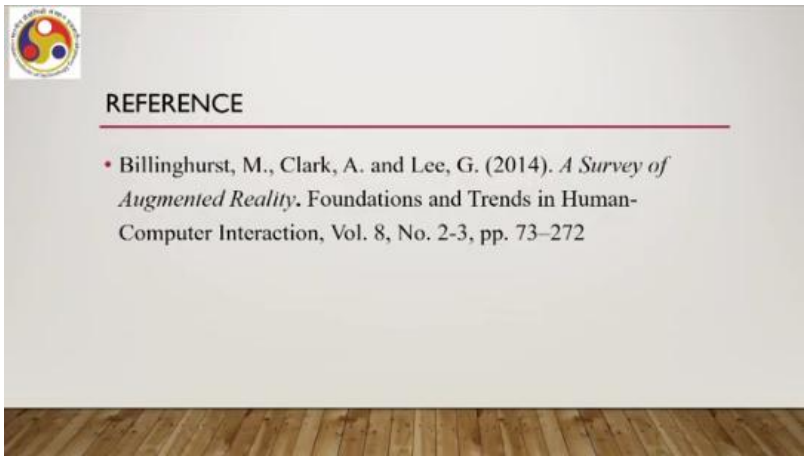
Users with basic PC skills could easily learn and use them, as coding is not required. So, useful for non-programmers who want to build AR applications without actually going through learning coding skills. However, as I said, absolute zero coding based applications will be limited in their utility as compared to some sort of advanced systems with scripting support, with scripts that will be providing more flexibility. An example of such standalone authoring tools is the Wikitude Studio.

It is a web-based authoring tool. It allowed users to create rich mobile AR content and also deploy either onto the Wikitude AR browser app or even create a custom mobile app. So, users can either develop a system and deploy it on a browser app or can build and deploy it as a mobile app. The studio is a package that comes with all these supports.

Now, this studio is no longer supported since last year. But they are still in use. The studio supported various types of media including 3D models and animations which can be registered to the real world. So that is one example of the full standalone content authoring tool or AR application authoring tool. which requires no coding skills.

There are many other such applications. So with that, we can conclude our discussion on AR software. So in summary, what we can say is that we have learned the basic idea of AR software. We have tried to understand the different tools and packages that are available in terms of four categories. The categories are based on programming skill required, at one extreme, we have the highest programming skill required when we are dealing with low-level libraries and frameworks, whereas on the other hand, at the other extreme, we require the least programming skill when we are dealing with standalone authoring tools. In

between, there are other categories such as rapid prototyping tools and one more category on plugins to existing systems. We have also gone through the basic idea of each of these categories with some examples to understand them to some extent. Next, in our next lecture, we will cover another important topic that is very much relevant to be able to fully understand, build, and deploy any, extended reality systems, be it virtual reality or augmented reality, that is where human factors affect our experience in these domains. So, we will discuss that by first discussing the idea of interactive systems, the role of human factors in interactive system building and deployment, and the factors that are most relevant for AR and VR experiences. So, I hope you have enjoyed the content, understood the topics covered, and learned about AR software. The topics that I have covered here are mostly taken from this reference. You are advised to go through this reference if you can access it. So with that, I would like to conclude this lecture. See you in the next lecture.



Thank you, goodbye, and happy learning.